VIT BHOPAL UNIVERSITY

GRAND HOTEL MANAGEMENT AND BILLING SYSTEM

Python Command-Line Project Report

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Academic Year 2026–27

Submitted To
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Python Programming

VITyarthi – Build Your Own Project

1. Introduction

Grand Hotel Management and Billing System is a Python-based command-line application developed as a practical programming project. It simulates a simple hotel booking and billing process in which a user enters customer information, selects accommodation and additional services, and receives a calculated final bill.

The project applies basic Python concepts such as input/output, strings, while loops, if/elif/else statements, validation and arithmetic operations. The system is intentionally kept simple so that each step of the program can be understood and tested directly from the terminal.

The project follows the VITyarthi Build Your Own Project requirement of applying course concepts to a meaningful real-world problem and documenting the design, implementation and evaluation.

Project Scope

The scope is limited to a single customer transaction during one program run. It covers customer details, room and bed selection, stay duration, food, spa services and GST-based billing. The current version does not store information permanently or connect to a database.

2. Problem Statement

Manual hotel billing can involve repeated calculations for accommodation and additional services. A simple computer program can reduce calculation effort by collecting choices through menus and automatically calculating the final amount.

The problem addressed by this project is to create a beginner-friendly hotel management and billing program that accepts valid customer inputs, provides service choices, handles invalid entries and produces a final bill with GST.

Objectives

• Create a simple command-line hotel booking workflow.

• Validate the phone number and menu selections.

• Provide room, bed, food and spa choices with fixed prices.

• Calculate charges according to the number of nights.

• Calculate subtotal, 18% GST and total payment.

• Display a clear final hotel bill.

3. Functional Requirements

The project contains more than three major functional modules, with clear input and output at each stage.

Module	Function
1. Customer Module	Accept username and validate a 10-digit phone number.
2. Accommodation Module	Select Standard, Deluxe or Suite room and Single, Double or King bed.
3. Stay Module	Accept and validate a positive number of nights.
4. Food Module	Select breakfast, lunch, dinner, all meals or no food.
5. Spa Module	Select head massage, body spa, full spa or no spa.
6. Billing Module	Calculate room, food, spa, subtotal, GST and total payment.
7. Output Module	Display the completed hotel bill and thank-you message.

4. Non-Functional Requirements

The following requirements describe qualities of the program rather than individual menu functions.

Requirement	Application in Project
Usability	Numbered menus and clear prompts make the program easy to follow.
Reliability	Validation loops prevent invalid phone numbers and out-of-range menu choices.
Performance	The program performs calculations immediately using a small number of operations.
Maintainability	Prices and choices are separated into clear conditional blocks, making basic changes straightforward.
Resource Efficiency	The program uses only standard Python input, output and arithmetic and does not require external packages.
Error Handling	Invalid entries trigger messages and repeated prompts instead of allowing the program to continue with invalid me

5. System Architecture

The application uses a simple sequential command-line architecture. The user provides input through the terminal. Validation checks the phone number and menu values. Valid selections are converted into prices, after which the billing block calculates the subtotal, GST and total.

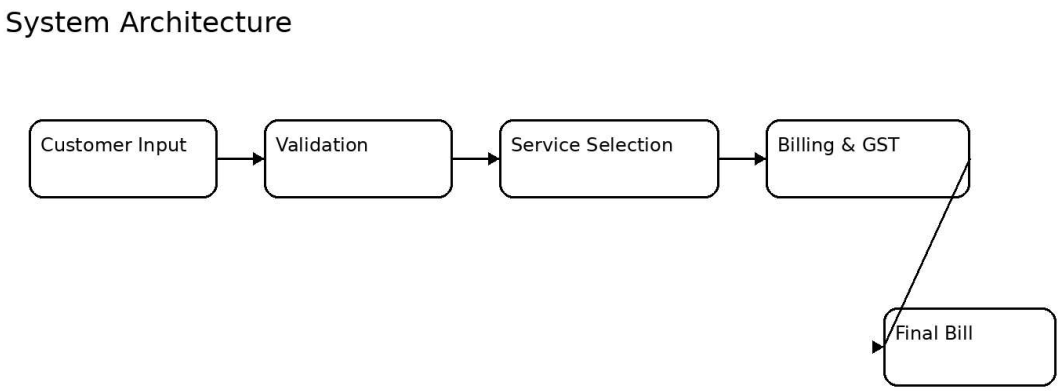


Figure 1. System architecture of the Grand Hotel application.

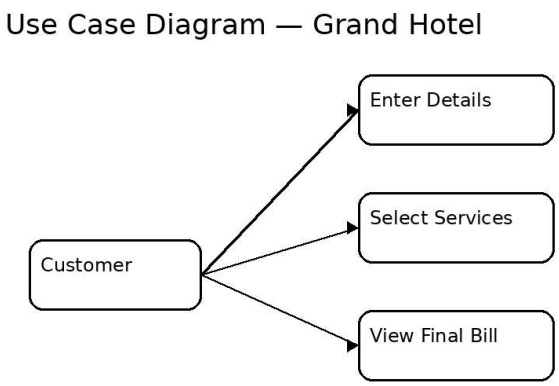


Figure 2. Use case diagram.

18. System Flowchart

End-to-end processing flow from customer details to final bill.

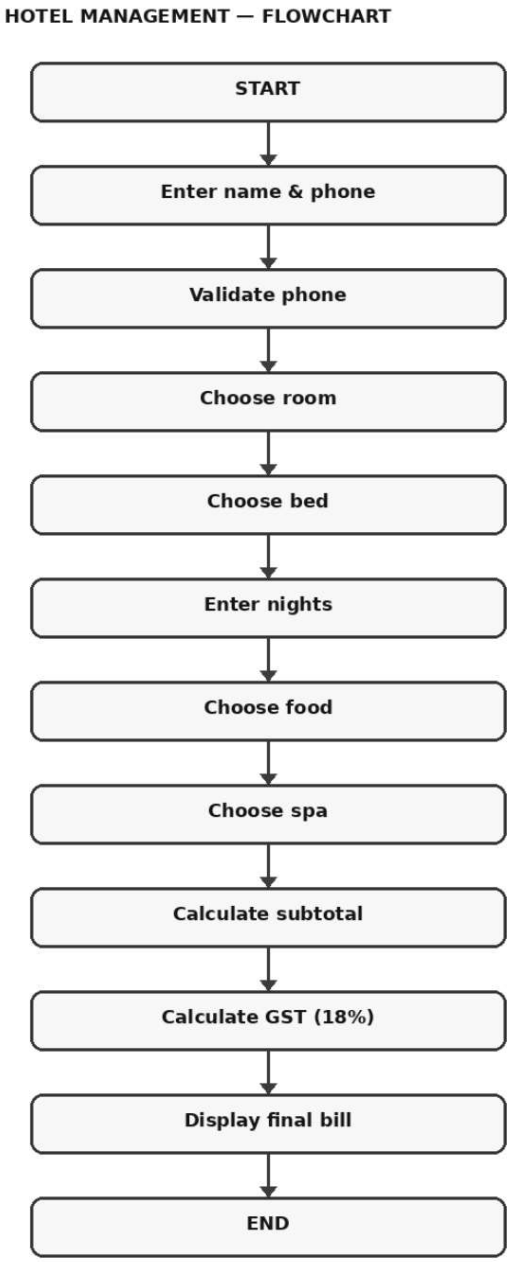


Figure 7. Flowchart of the HOTEL MANAGEMENT application.

Figure 3. Workflow/process flow used for the project.

6. Design Diagrams

Use Case Diagram

Workflow Diagram

Sequence Diagram

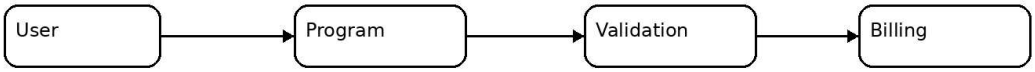


Figure 4. Sequence diagram.

Component Diagram

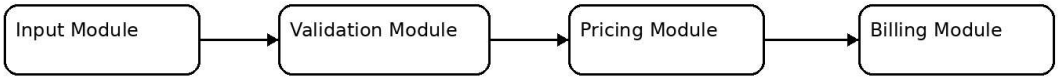


Figure 5. Component view.

6. Design Diagrams — Continued

Sequence Diagram

Class / Component Diagram

ER Diagram

Not applicable. The current program does not use a database or permanent storage, so there is no ER diagram or database schema.

7. Design Decisions & Rationale

Python was selected because the project is based on introductory programming concepts such as variables, strings, conditions, loops and arithmetic calculations.

A command-line interface was selected because it keeps the application easy to execute and demonstrates the program logic directly without requiring a graphical framework.

Numbered menus were used for room, bed, food and spa choices so that the user can make selections easily. While loops were used for repeated validation so invalid values are not accepted as valid menu selections.

The project uses fixed prices because the objective is to demonstrate programming logic and billing calculations rather than real-time hotel inventory or pricing.

The current version does not use a database because the supplied program is designed for one transaction during one execution. Future versions can introduce storage if the project scope is expanded.

The billing formula is deliberately transparent: room and bed charges are multiplied by nights, food is multiplied by nights, spa is charged once, then 18% GST is added to the subtotal.

8. Implementation Details

The final program is written as one sequential Python file, **grand_hotel.py**. It does not require third-party libraries.

Pricing Used

Category	Options and Price
Room	Standard ■2000 | Deluxe ■3500 | Suite ■5000
Bed	Single ■100 | Double ■500 | King Size ■1000
Food	Breakfast ■300 | Lunch ■500 | Dinner ■500 | All Meals ■1100 | No Food ■0
Spa	Head Massage ■500 | Body Spa ■1000 | Full Spa ■1500 | No Spa ■0
Tax	GST = 18% of subtotal

9. Screenshots / Results — Full Source Code

The following is the actual final source code rendered for the project. It is split into two pages so the code remains readable.

Programiz Python — grand_hotel.py — FULL CODE (Part 1/2)

```
print("WELCOME TO THE GRAND HOTEL")

name=input("PLEASE ENTER YOUR USER NAME: ")
phone=input("PLEASE ENTER YOUR PHONE NUMBER: ")

while len(phone)!=10 or not phone.isdigit():
  print("INVALID PHONE NUMBER! PLEASE ENTER A 10-DIGIT NUMBER.")
  phone=input("PLEASE ENTER YOUR PHONE NUMBER: ")

print("\nCHOOSE YOUR ROOM: ")
print("1. STANDARD ROOM")
print("2. DELUXE ROOM")
print("3. SUITE ROOM")

room=input("ENTER YOUR CHOICE: ")

while room=="" or not room.isdigit() or int(room)<1 or int(room)>3:
  print("INVALID CHOICE! PLEASE CHOOSE 1,2 OR 3.")
  room=input("ENTER YOUR CHOICE: ")

room=int(room)

if room==1:
  room_price=2000
  print("STANDARD ROOM SELECTED.")
elif room==2:
  room_price=3500
  print("DELUXE ROOM SELECTED.")
else:
  room_price=5000
  print("SUITE ROOM SELECTED.")


print("\nCHOOSE BED SIZE: ")
print("1. SINGLE BED")
print("2. DOUBLE BED")
print("3. KING SIZE BED")

bed=input("ENTER YOUR CHOICE: ")

while bed=="" or not bed.isdigit() or int(bed)<1 or int(bed)>3:
  print("INVALID CHOICE! PLEASE CHOOSE 1,2 OR 3.")
  bed=input("ENTER YOUR CHOICE: ")

bed=int(bed)

if bed==1:
  bed_price=100
  print("SINGLE BED SELECTED.")
elif bed==2:
  bed_price=500
  print("DOUBLE BED SELECTED.")
else:
  bed_price=1000
  print("KING SIZE BED SELECTED.")


nights=input("\nENTER NUMBER OF NIGHTS: ")

while nights=="" or not nights.isdigit() or int(nights)<=0:
  print("INVALID NUMBER OF NIGHTS!")
  nights=input("ENTER NUMBER OF NIGHTS: ")

nights=int(nights)

room_total=(room_price+bed_price)*nights


print("\nCHOOSE FOOD: ")
print("1. BREAKFAST - Rs. 300")
print("2. LUNCH - Rs. 500")
```

Figure 6. Full source code — first section.

9. Screenshots / Results — Full Source Code Continued

Programiz Python — grand_hotel.py — FULL CODE (Part 2/2)

```
print("3. DINNER - Rs. 500")
print("4. ALL MEALS - Rs. 1100")
print("5. NO FOOD")

food=input("ENTER YOUR CHOICE: ")

while food=="" or not food.isdigit() or int(food)<1 or int(food)>5:
  print("INVALID CHOICE! PLEASE CHOOSE 1,2,3,4 OR 5.")
  food=input("ENTER YOUR CHOICE: ")

food=int(food)

if food==1:
  food_price=300
elif food==2:
  food_price=500
elif food==3:
  food_price=500
elif food==4:
  food_price=1100
else:
  food_price=0

food_total=food_price*nights


print("\nCHOOSE SPA SERVICE: ")
print("1. HEAD MASSAGE - Rs. 500")
print("2. BODY SPA - Rs. 1000")
print("3. FULL SPA - Rs. 1500")
print("4. NO SPA")

spa=input("ENTER YOUR CHOICE: ")

while spa=="" or not spa.isdigit() or int(spa)<1 or int(spa)>4:
  print("INVALID CHOICE! PLEASE CHOOSE 1,2,3 OR 4.")
  spa=input("ENTER YOUR CHOICE: ")

spa=int(spa)

if spa==1:
  spa_price=500
elif spa==2:
  spa_price=1000
elif spa==3:
  spa_price=1500
else:
  spa_price=0

spa_total=spa_price


subtotal=room_total+food_total+spa_total
gst=subtotal*18/100
total=subtotal+gst


print("\nHOTEL BILL")
print("GRAND HOTEL")
print("\nCUSTOMER NAME: ",name)
print("PHONE NUMBER: ",phone)
print("ROOM CHARGES: Rs",room_total)
print("FOOD CHARGES: Rs",food_total)
print("SPA CHARGES: Rs",spa_total)
print("SUBTOTAL: Rs.",subtotal)
print("GST(18%): Rs",gst)

print("\nTOTAL PAYMENT: Rs",total)

print("\nTHANK YOU FOR VISITING GRAND HOTEL!")
print("HAVE A NICE DAY!")
```

Figure 7. Full source code — remaining section.

9. Screenshots / Results — Welcome and Main Options

This screenshot shows the welcome message, customer entry and the room and bed menus from an execution of the final program.

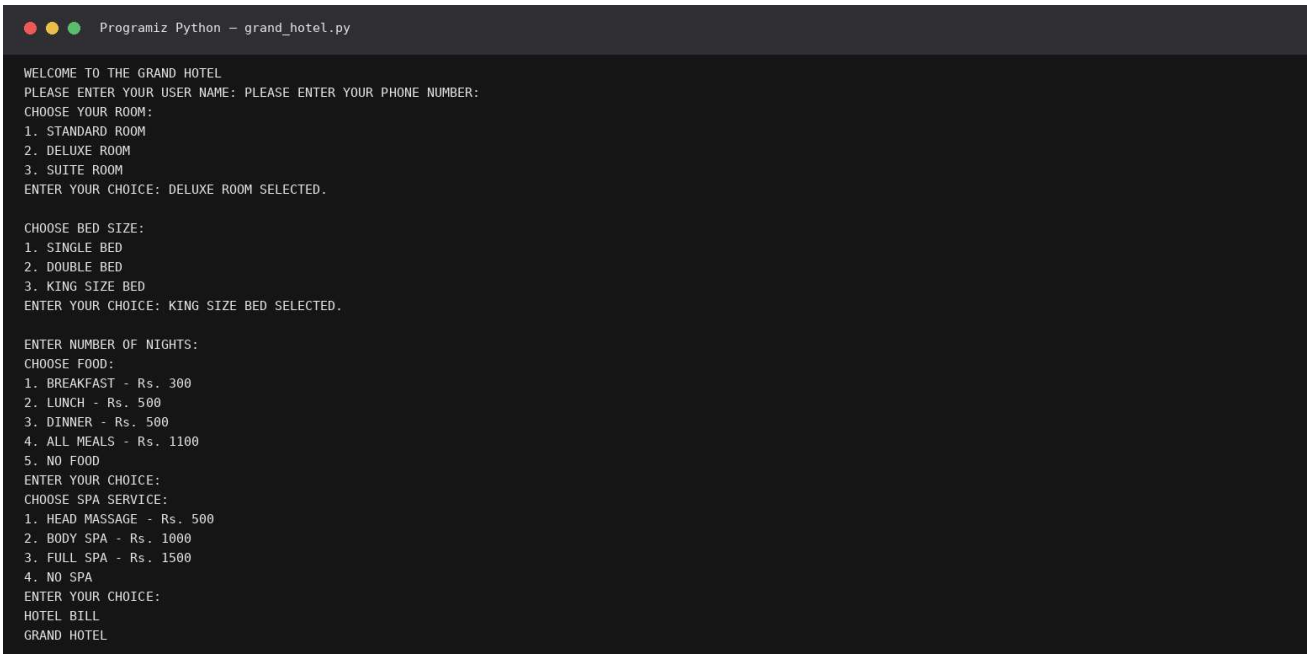


Figure 8. Welcome to Grand Hotel, customer details, room and bed options.

9. Screenshots / Results — Remaining Options

This screenshot shows the remaining options after accommodation: number of nights, food and spa services.

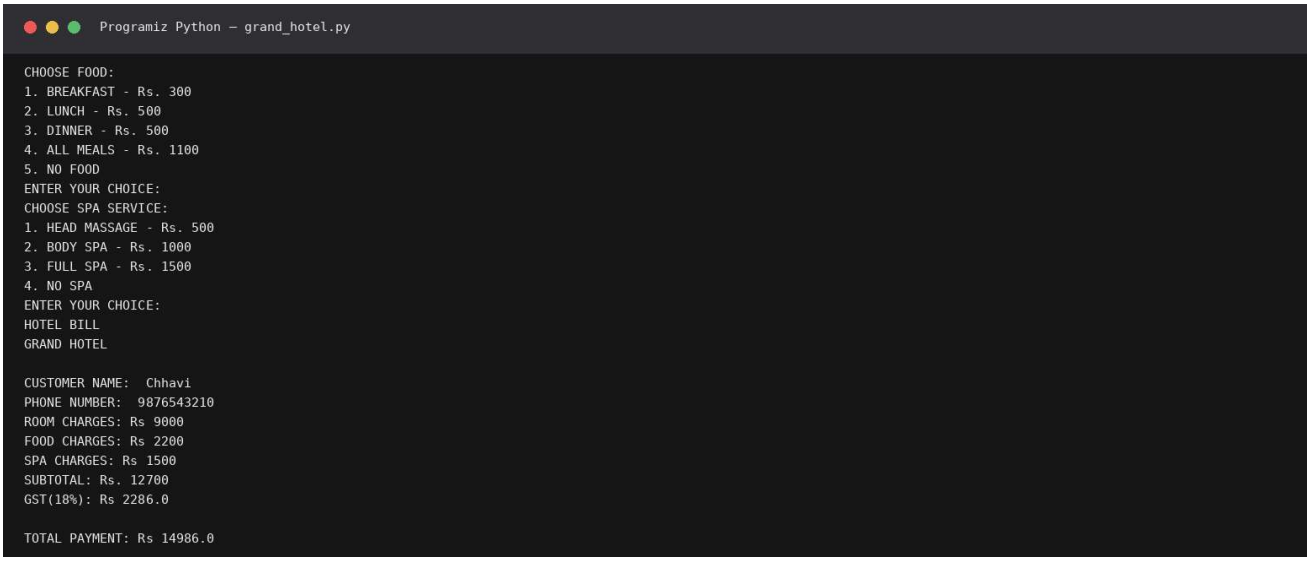


Figure 9. Number of nights, food and spa options.

9. Screenshots / Results — Error Handling

The following run intentionally contains invalid inputs. The validation loops display an error message and request the input again before continuing.

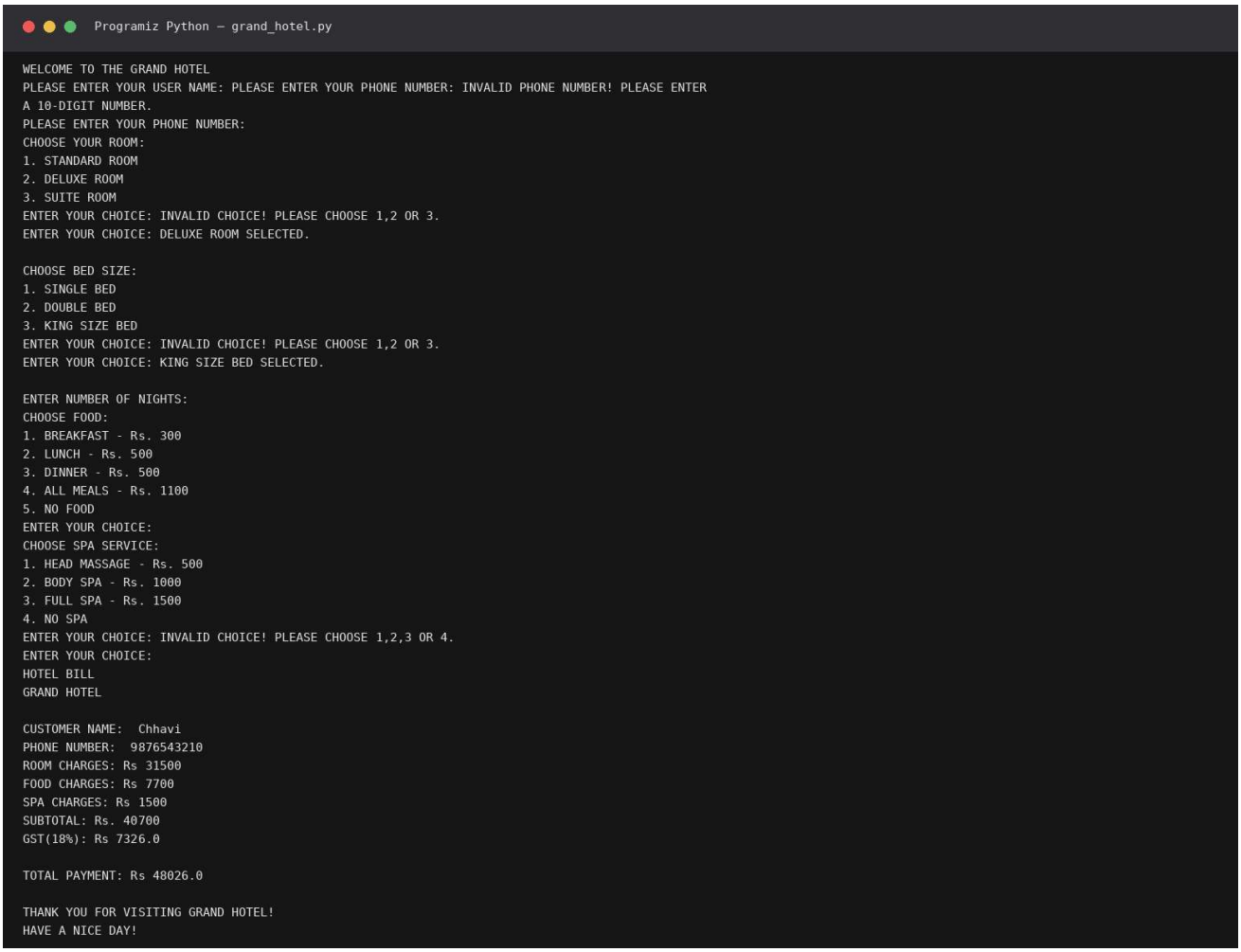


Figure 10. Error handling and repeated input validation.

9. Screenshots / Results — Final Billing

The final billing screenshot is from the same project logic. It shows the calculated room, food and spa charges, subtotal, 18% GST and final payment.



Figure 11. Final hotel bill and total payment.

10. Testing Approach

Testing was carried out using valid and invalid input scenarios. The main objective was to confirm that validation loops work and that the billing formula produces the expected result.

Test	Scenario	Expected Result
TC-01	Phone has fewer than 10 digits	Invalid message and re-entry
TC-02	Room choice outside 1–3	Invalid message and re-entry
TC-03	Bed choice outside 1–3	Invalid message and re-entry
TC-04	Nights = 0	Invalid message and re-entry
TC-05	Food choice outside 1–5	Invalid message and re-entry
TC-06	Spa choice outside 1–4	Invalid message and re-entry
TC-07	All valid selections	Final bill generated

11. Challenges Faced

• Handling incorrect phone numbers and repeated input.

• Validating menu values while keeping the program beginner-friendly.

• Making room and food charges depend on the number of nights.

• Keeping the billing calculation simple and readable.

• Preparing readable evidence screenshots of both code and execution.

12. Learnings & Key Takeaways

• Learned how to use while loops for input validation.

• Practised if/elif/else statements for menu-based choices.

• Used string methods such as len() and isdigit() for validation.

• Applied arithmetic expressions to a practical billing problem.

• Learned how to test both valid and invalid program inputs.

• Learned how program design, documentation, screenshots and testing support a coding project.

13. Future Enhancements

• Add room availability and booking IDs.

• Store customer and booking history using CSV or a database.

• Support multiple rooms or multiple customers.

• Add dates and check-in/check-out information.

• Add a graphical user interface.

• Generate a printable invoice or PDF bill.

15. References

1. Vidyarthi, "Build Your Own Project – General Project Instructions & Submission Guidelines," supplied course document.

2. Python 3 documentation and course material used for basic syntax, input/output, conditions, loops and string methods.

3. Grand Hotel Management and Billing System source code, **grand_hotel.py**, included with this project package.

4. Project screenshots and design diagrams included in this report.

Conclusion

The Grand Hotel Management and Billing System demonstrates how basic Python programming concepts can be applied to a practical hotel billing problem. The project provides a clear input-output workflow, validation, service selection and automated GST-based billing while remaining simple enough for a first-semester programming project.